# Alexander Bernstein

alexbstl@gmail.com 314-283-0026 alexbstl alexbstl

## Education

**University of California, Santa Barbara** Santa Barbara, CA
*Ph.D. in Statistics* *Sept. 2025*
Dissertation: "Long-Only Minimum Variance Portfolio Composition for Factor Models"

**Washington University in St. Louis** St. Louis, MO
*M.Sc. in Systems Science and Applied Mathematics* *Dec. 2016*
*B.A. in Mathematics and Economics (Cum Laude)* *May 2014*

## Technical Skills

**Languages & Tools:** Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, CVXPY), SQL, R, C, Java, JavaScript/NodeJS, Git, Linux, AWS, MongoDB, LaTeX.

**Machine Learning:** Neural Networks, Gaussian Processes, Reinforcement Learning (Q-Learning), Ensemble Methods (Boosting/Bagging), Bayesian Inference, Time Series Forecasting, Feature Engineering.

**Data & Modeling:** Statistical Modeling, Experimentation, High-Dimensional Regression, Optimization, Backtesting & Simulation, Data Pipelines.

## Selected Projects

*End-to-End Equity Modeling Pipeline:* Architected a modular, spec-driven (**YAML**) **Python** pipeline for S&P 500 portfolio construction: pluggable covariance/return estimators via decorator registries, **Numba**-JIT Kalman filters, **CVXPY** optimization, a walk-forward backtest engine, and a CLI rebalancer that emits live trades. Built on clean ABCs with unit tests and reproducible configs. **[Add: data volume / runtime.]**

*ML Surrogates for Derivative Pricing:* Trained **Gaussian Process** (Matérn) and **PyTorch** neural-network surrogates to emulate Monte Carlo and finite-difference (Crank-Nicolson) solvers for the **Heston** and **CEV** models, recovering prices and **Greeks** (via autograd) at a fraction of the numerical cost.

*Conditional Covariance Estimation:* Built a **Covariance Regression** model (**CovRegpy**) for covariate-adjusted covariance of idiosyncratic equity returns — a scalable alternative to **MGARCH** that sharply reduces parameter count in high dimensions.

*Multi-Agent Reinforcement Learning:* Implemented a **Q-Learning** agent for a mean-field cyber-security environment, learning robust defense policies against a non-stationary population distribution $\mu$.

## Experience

**University of California, Santa Barbara** Santa Barbara, CA
*Teaching Assistant & Graduate Student Mentor* *Sept. 2017 – June 2025*
Taught statistics, probability, regression, time series, and stochastic processes; mentored undergraduate research projects in modeling and optimization, including posters and written reports.

**Epic Systems** Madison, WI
*Technical Services Engineer* *Sept. 2014 – Sept. 2015*
Diagnosed customer issues and shipped improvements to the Epic EHR codebase for large institutional clients under high-availability and regulatory constraints.

**Prozess Technologie** St. Louis, MO
*Intern* *May 2013 – May 2014*
Built computational simulations to validate laboratory equipment and calibrated pharmaceutical manufacturing hardware to industry standards.

## Research & Publications

Banerjee, T., **Bernstein, A.**, Feinstein, Z. (2025). "Dynamic clearing and contagion in financial networks." *European Journal of Operational Research* 321(2), 664–675.

**Bernstein, A.**, Shkolnik, A. (2025). "Asymptotics of Quadratic Forms on a Simplex." *In preparation.*